$p + p \to \pi^+\pi^-e^+e^-$ Analysis Report

Short summary of Part I: simulation

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Summary of the full report `report.pdf` (45 pages), where every number is derived and documented. Simulation only: no measured data and no combinatorial background (CB). Expected numbers of rows in data use $L_{\text{data}} = 6080.5\,\text{nb}^{-1}$, which is **not yet confirmed**.

Table 1: Notation used in this summary, in the order of appearance. The full report defines every item in detail (its "Notation and abbreviations" table points to the sections).

Samples and normalisation	
SMASH, PLUTO, HGeant	event generator; decay generator used by the converter; GEANT3-based detector simulation of HADES
DST, PAT, FAT	reconstructed data; PostDST Analysis Tool (hypotheses, PID, ranking); Final Analysis Tool (selection, kinematics)
T, $b_{\max}$, N	beam kinetic energy; largest impact parameter of SMASH; number of simulated events per set
L_{sim}, L_{data}	integrated luminosity of one simulated set ($221.05\,\text{nb}^{-1}$); of the data ($6080.5\,\text{nb}^{-1}$, not confirmed)
std, lep, η set	the three simulation sets (natural decays; dilepton decays forced; η decays forced)
row	one combination π^+ π^- e^+e^- of reconstructed tracks; an event can give several rows
isBest	PAT ranking of the combinations of an event (1 = best, 0 = others); "all combinations" = isBest ≥ 0
PT3	the physics trigger used for this analysis (bit 8192 of the trigger word)
"pre"	base selection: isBest ≥ 0, vertex $z > -500\,\text{mm}$, PT3, vertex $z > -160\,\text{mm}$ (target), corrected momenta
Cross sections and counts	
σ_{prod}	production cross section of a particle in SMASH (mean number per collision $\times\, \pi b_{\max}^2$)
BR	branching ratio (PDG values unless stated)
ε	acceptance $\times$ efficiency: reconstructed rows after "pre" per generated decay
σ_{vis}	visible cross section of rows after "pre", $\sigma_{\text{prod}} \times \text{BR} \times \varepsilon$; rows at $L_{\text{data}} = \sigma_{\text{vis}} L_{\text{data}}$
bump, peak	$\eta \to \pi^+\pi^-\pi^0$ seen below m_η (photon lost); η seen at m_η ($\eta \to \pi^+\pi^-e^+e^-$ or $\eta \to \pi^+\pi^-\gamma$ with conversion)
conversion	a photon converted into an e^+e^- pair in the detector material
X17	hypothetical boson of $17\,\text{MeV}/c^2$ decaying into e^+e^-; $\text{BR} = \text{BR}(\eta \to \pi^+\pi^- X17)\,\text{BR}(X17 \to e^+e^-)$
physical background	rows with a true e^+e^- pair from one source (π^0/η Dalitz decay, conversion, ρ, ω, ϕ) and any pions
multi-pion	physical background from π^0 Dalitz and conversion pairs with pions (no η)
fake lepton; two sources	a lepton slot holds a non-lepton or a ghost track; the two leptons come from different decays
CB	combinatorial background of the data (absent in simulation)
Corrections	
`geninfo`	generator code of a GEANT track (decay channel), stored by PAT as a 32-bit float
rules v2	weight rules with the exact HGeant branching ratios (from the HGeant card `ETA`)
QED (Kroll–Wada), form factor	physical distribution of $M(e^+e^-)$ in a decay through a virtual photon; its hadronic modification
OA	opening angle between e^+ and e^- (lab)
Kinematic variables and cuts	
$M(\pi^+\pi^-e^+e^-)$, $M(e^+e^-)$, $M(\pi^+\pi^-)$	invariant mass of $\pi^+\pi^-e^+e^-$, of the e^+e^- pair, of the $\pi^+\pi^-$ pair

MM_4	missing mass of $\pi^+\pi^-e^+e^-$: mass of (beam + target − π^+ π^- e^+e^-) four-momenta; for $pp \to pp\eta$ it is $M(pp)$
α_{lab}	angle between the momenta of the $\pi^+\pi^-$ and the e^+e^- systems (lab)
α_η	angle between $\pi^+\pi^-$ and e^+e^- in the rest frame of a hypothetical η with the measured $\pi^+\pi^-e^+e^-$ momentum and mass m_η
Δz_π	largest distance in z between a pion track (closest approach to the beam axis) and the event vertex
pions in the vertex fit	both pion tracks were used in the fit of the event vertex (PAT flag `isUsedVertex`)
peak / bump window	$0.5333 \le M(\pi^+\pi^-e^+e^-) < 0.5587$ / $0.4576 \le M(\pi^+\pi^-e^+e^-) < 0.5255\,\mathrm{GeV}/c^2$
S, B	expected signal rows and other (physical-background) rows in a window at L_{data}
purity, efficiency	$S/(S+B)$; fraction of the signal of "pre" kept by the cuts
F	figure of merit $S/\sqrt{S+B}$
Test of the extraction	
half A / B	the simulated events split in two (even / odd event-group number): templates from A, pseudo-data from B
toy	one pseudo-data set drawn from half B (Poisson bootstrap)
bias; pull	mean (extracted − true) over the toys; (extracted − true)/quoted error
coverage	fraction of toys whose quoted error contains the truth (0.68 expected)
UL	expected 90 % CL upper limit on the X17 branching ratio

1 Samples and normalisation

- One SMASH run, p+p at $T = 4.5\,\mathrm{GeV}$, 10^{10} events, impact parameter uniform in the area up to $b_{\max} = 1.2\,\mathrm{fm}$ (checked in the SMASH configuration and output). Each event stands for $\pi b_{\max}^2/N$, so every set has $L_{\mathrm{sim}} = 221.05\,\mathrm{nb}^{-1}$.
- Three sets of the *same* events (event by event identical generator content; HGeant rotates each event by a random azimuth): **standard** (all decays natural), **lepton** (every π^0, η, ρ^0, ω, ϕ forced into a dilepton channel), η (every η forced: 50 % $\eta \to \pi^+\pi^-e^+e^-$, 50 % $\eta \to \pi^+\pi^-X17$).
- Chain: HGeant, DST (feb22, gen4), PAT (PostDST Analysis Tool, simulation PID cuts) and FAT (Final Analysis Tool). Base selection "pre": all combinations (isBest ≥ 0), reconstructed vertex $z > -500\,\mathrm{mm}$, PT3 trigger, target region $z > -160\,\mathrm{mm}$, momenta corrected for energy loss.

2 Channels, cross sections and expected counts

All cross sections here are *visible* cross sections of reconstructed rows after "pre": $\sigma_{\mathrm{vis}} = \sigma_{\mathrm{prod}} \times \mathrm{BR} \times \varepsilon$, with the production from the SMASH model, the branching ratios from PDG and the acceptance times efficiency ε from the HADES simulation.

Table 2: Signal channels (notation: Table 1). $\sigma_{\mathrm{prod}}(\eta) = 1.0665 \pm 0.0016\,\mathrm{mb}$ (SMASH: 0.0236 η per collision). ε: reconstructed rows per generated decay after "pre". σ_{vis} with the physical decay kinematics (QED and form factors for the direct decay, measured Dalitz plot of $\eta \to \pi^+\pi^-\pi^0$, pion form factor for $\eta \to \pi^+\pi^-\gamma$). Rows: expected at L_{data}, all $M(\pi^+\pi^-e^+e^-)$. X17 for an illustrative BR $= 10^{-5}$.

	Process	BR (PDG)	ε	σ_{vis} [nb]	rows at L_{data}
bump	$\eta \to \pi^+\pi^-\pi^0$, $\pi^0 \to e^+e^-\gamma$	0.2292×0.01174	2.5×10^{-3}	7.05 ± 0.20	42 900
bump	$\eta \to \pi^+\pi^-\pi^0$, one π^0 photon converts	0.2292×0.98823	2.1×10^{-5}	5.19 ± 0.17	31 600
peak	$\eta \to \pi^+\pi^-\gamma$, the photon converts	0.0422	2.8×10^{-5}	1.24 ± 0.07	7 500
peak	$\eta \to \pi^+\pi^-e^+e^-$ (direct)	2.68×10^{-4}	5.0×10^{-3}	1.417 ± 0.003	8 600
X17	$\eta \to \pi^+\pi^- X17$, $X17 \to e^+e^-$	10^{-5} (hyp.)	6.7×10^{-3}	0.0712	430
physical background (true e^+e^- pair from one source, any pions)				686	4.17×10^6
of which multi-pion (π^0 Dalitz and conversion pairs): 94 %					
fake lepton / leptons from two sources (shown, not added)				125 / 12.5	

The simulated spectrum after "pre" (Fig. 1) is dominated by the multi-pion background; the η structures sit on top of it. Conversions are 42 % of the simulated bump and almost half of the peak.

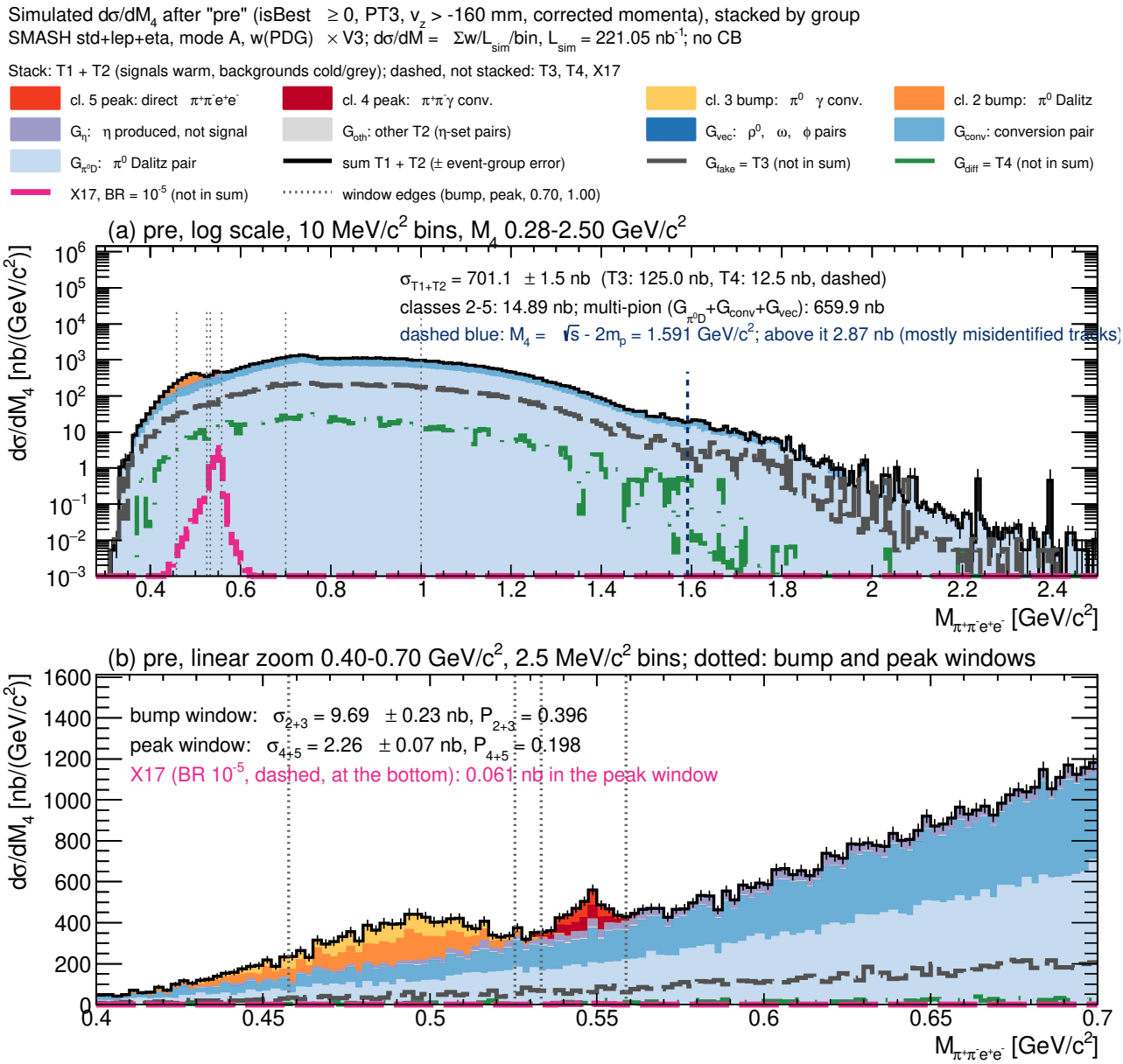


Figure 1: Simulated $d\sigma/dM(\pi^+\pi^-e^+e^-)$ after "pre", stacked by component: signal classes (warm colours) and background groups (cold colours). Dashed, not stacked: fake leptons, leptons from two sources and X17 at BR $= 10^{-5}$.

3 What was corrected

1. **Generator codes stored as 32-bit floats.** Codes above 2^{24} are rounded in the PAT ntuples (e.g. $\eta \to \pi^+\pi^-e^+e^-$: 1 702 030 809 is stored as 1 702 030 848). All values were mapped back unambiguously; a cut `TTree::Draw("x==1702030809")` selects nothing.

2. **Weights of the lepton set.** Same decay = equal (`geninfo`, `geninfo2`), not the triple of the producers' note (that rule weights a Dalitz lepton paired with its own conversion lepton 83 times too low); the weight of a row is the product over the distinct forced decays of all four tracks (leptons also occur in pion slots); the converter's second-decay weight needs a factor 0.0985.

3. **Ownership between sets.** Every row type is taken from one set only; the sets agree where they overlap (non-forced rows of the η set against the standard set: 1.0056 ± 0.0036).

4. **HGeant η decay table.** The HGeant input of the production sets BR($\eta \to e^+e^-\gamma$) = 6.0 %

instead of 0.60 % (card `ETA 37.30 30.20 21.00 4.75 6.0 7.1e-2`). Weights now use the exact card values (rules v2): bump −1.7 %, conversion peak +3.9 %.

5. **The $\eta \to \pi^+\pi^-e^+e^-$ generator.** The channel is defined without a dilepton (virtual photon), so its $M(e^+e^-)$ is not QED (27 % of the pairs within 0.5 MeV/c^2 of threshold, 4.7 % above 20 MeV/c^2 against 28 % in QED). Rows are reweighted to Kroll–Wada with form factors; the cut OA$> 2°$ then keeps 0.70 of the simulated peak (0.59 before; data 0.78). The PLUTO plugin `eta_physics` stays off on purpose; the producers should define the channel as `eta:dilepton[e+,e-],pi+,pi-`.

6. **Bookkeeping.** Simulation PT3 rows have `trigbit` 12288 (bit test needed); the merged files contain several key cycles (use the highest); rows are grouped into events by adjacency; weighted sums in double precision (`TH1F` lost 3.4 %).

4 Final selection

The cuts were optimised on simulation truth with $F = S/\sqrt{S+B}$, an exhaustive scan of nine cut variables, a rule fixed in advance (fewest cuts within 5 % of the best F on the full sample and on both halves) and a split-half check of the optimisation bias (below 1 % for the chosen cuts).

Table 3: Frozen selections (all in addition to "pre"; notation: Table 1). S, B: signal and background rows in the window at L_{data}. Δz_π: largest distance in z of the two pion tracks from the event vertex.

Target	Cuts	S	B	purity	efficiency
peak (direct + conversion), $0.5333 \le M(\pi^+\pi^-e^+e^-) < 0.5587$	$\Delta z_\pi < 40$ mm, $\mathrm{MM}_4 < 2.85$ GeV/c^2	12 800	41 800	0.23	0.93
direct decay only, same window	$\Delta z_\pi < 40$ mm	7 100	52 500	0.12	0.96
bump, $0.4576 \le M(\pi^+\pi^-e^+e^-) < 0.5255$	none	58 900	89 800	0.40	1
X17, peak window	$15.5 \le M(e^+e^-) < 24.5$ MeV/c^2, both pions in the vertex fit	213 at 10^{-5}	5 340		

- The optimum is flat: many selections are within a few per cent of the best F. The gain of the frozen peak cuts over "pre" is 5 % in F.
- The usual η cuts (OA$> 2°$, α_{lab}, $M(\pi^+\pi^-)$, α_η) remove background *outside* the windows but do not improve them; OA$> 2°$ costs 19 % of the direct decay. In data they may still pay off against the CB, which the simulation does not contain: to be tested on data.
- Δz_π is not yet available in the data compact rows; the data-ready equivalent on the plateau is "both pions in the vertex fit".
- X17: expected 90 % CL limit 1.9×10^{-5} (at L_{data}), set by the simulation statistics of the background; 5.7×10^{-6} if the background were known exactly.

5 Signal extraction compared with the truth

The methods planned for data were applied blind to pseudo-data: half of the simulated events gave the templates, the other half 200 Poisson-bootstrap toys (and 1500 parametric toys for X17); the true content was revealed only after the procedure was frozen. Fig. 2 shows one toy with the fits, Table 4 and Fig. 3 the result.

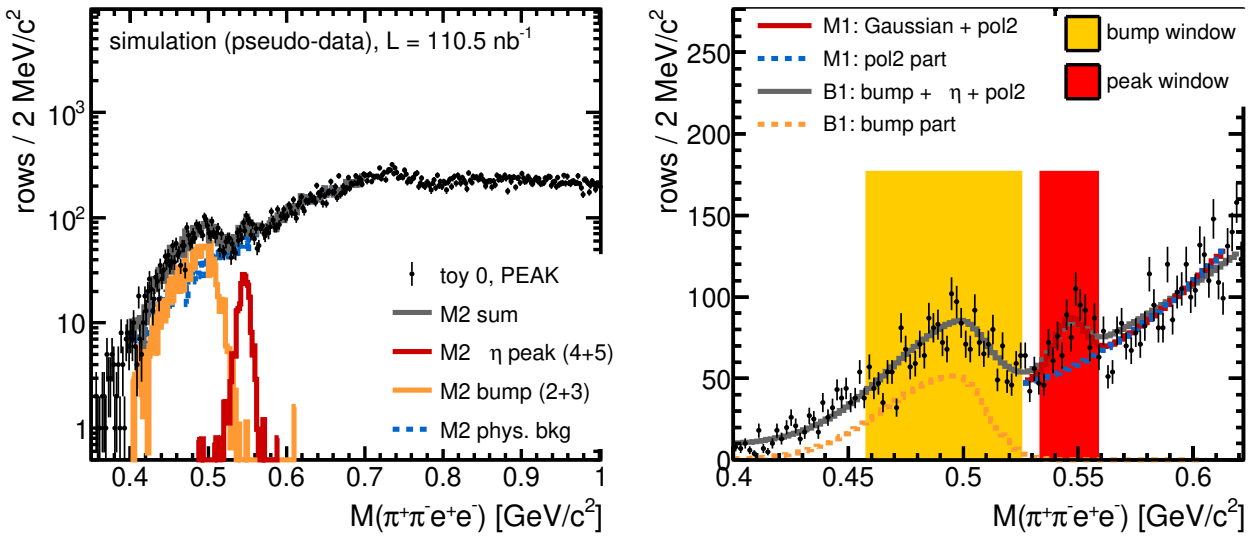


Figure 2: One pseudo-data toy (110.5 nb^{-1}, peak selection). Left: $M(\pi^+\pi^-e^+e^-)$ with the components of the template fit. Right: zoom with the Gaussian + polynomial fit of the peak and the parametric fit of the bump, with the bump and peak windows.

Table 4: Extracted yield against the truth, mean over 200 toys (notation: Table 1). The common uncertainty of the simulation templates (±11–15 % peak, ±4.5–5 % bump) is shared by all methods. Coverage: fraction of toys within the quoted error (nominal 0.68).

Yield	Method	bias	coverage	verdict
peak	template fit	+1.4 %	0.77	good
	Gaussian + polynomial, calibrated	+3.1 %	0.72	good
	linear side bands	−21 %	0.53	loses signal: do not use
bump	side band with simulated shape	+3.9 %	0.66	good
	edge step	+1.3 %	0.76	unbiased, 16 % error
	template fit / parametric fit	+7.7 / +5.4 %	0.31 / 0.66	somewhat high
direct decay alone	template fit, conversion constrained	+21 %	0.62	limited by the conversion template

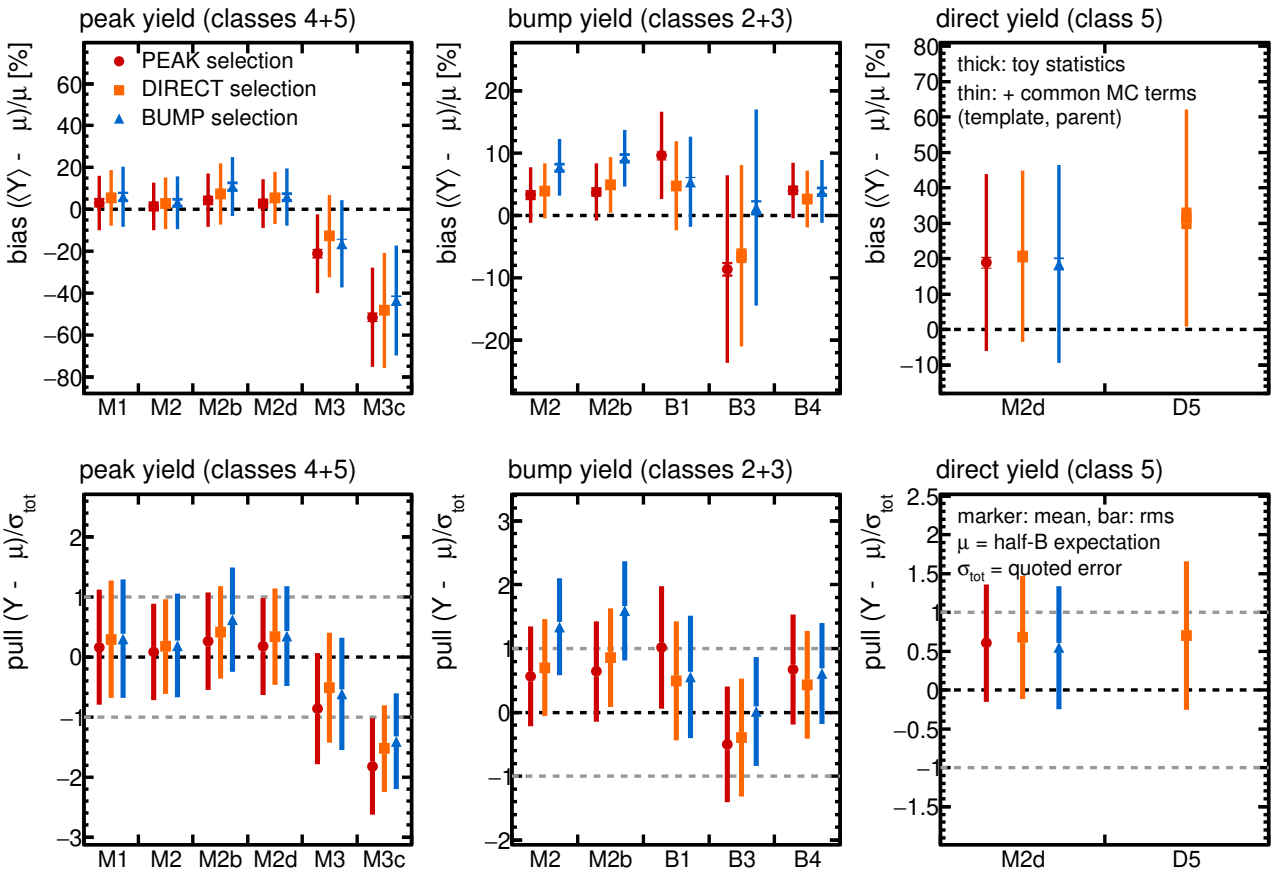


Figure 3: Blind test, η yields. Top: relative bias per method and selection (thick bars: toy statistics, thin bars: including the common simulation error). Bottom: pull, marker = mean, bar = rms.

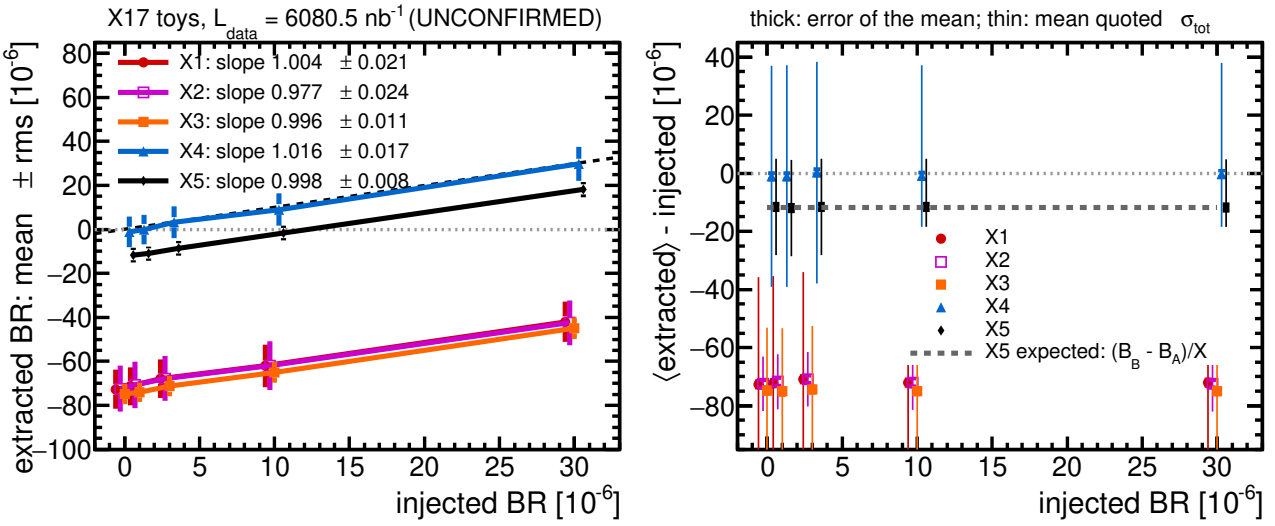


Figure 4: Blind test, X17: extracted against injected branching ratio (300 toys per point) with straight-line fits, and the offset at zero. All estimators follow the injected value with slope 0.98–1.02; the common offset comes from the background of the simulation sample.

What follows.

- **Peak:** the template fit and the calibrated Gaussian fit reproduce the truth within a few per cent; in a single data set they may differ from each other by about 12 %. Linear side bands lose 20–45 % because the background is convex under the peak.

- **Bump:** the side band with the simulated background shape is closest; the edge step is unbiased but imprecise.
- **Direct decay alone:** $M(\pi^+\pi^-e^+e^-)$ cannot separate it from the conversion peak; the split is limited by the 307 simulated rows of the conversion channel.
- **X17:** the response is linear; a valid limit must include the uncertainty of the background, otherwise a background fluctuation can fake a signal (Fig. 4).
- The quoted errors describe the spread of the toys; the common simulation error must be added.

6 Open points

- L_{data} to be confirmed; no CB in this study (the cut choice must be repeated with data).
- The standard set limits the bump, conversion and X17-background templates: a larger production would reduce the dominant simulation error.
- Producers: confirm the HGeant $\eta \to e^+e^-\gamma$ setting (6.0 %) and regenerate the η set with a dilepton definition of $\eta \to \pi^+\pi^-e^+e^-$.
- Beam energy: the analysis uses $T = 4530\,\text{MeV}$, SMASH 4500 MeV (affects the missing mass only).